

frames, which were strong, light, and easier to make. On aircraft such as the Albatros D-series fighters, they took plywood and wrapped it in strips around an inner framework to create wooden-skinned monocoque fuselages.

Although the Albatros fighters were fine machines, their evolution showed the increasing difficulties the Germans ran into, trapped between the demands of quantity and quality. Aiming to achieve its 1,000-aircraft-a-month target for 1917, Germany opted for mass production of the Albatros D.V, a variant of the D.III. But by the summer of that year, the Albatroses were being outclassed by a new generation of Allied fighters, especially the S.E.5 and the SPAD XIII. Yet dedicated to meeting their production targets, German factories went on churning out Albatroses into 1918.

The skies over the Western Front were an essentially Darwinian environment in which aircraft constantly evolved to survive – sometimes by straight imitation. When the French Nieuport 17 scout threatened the predominance of the German Albatros D.II fighter in 1916, for example, the Germans simply copied the Nieuport's single-spar lower wing – and with it, its tendency to twist and fail – to create the Albatros D.III. Similarly, after the Sopwith Triplane flown by Britain's Royal Naval Air Service shocked German fliers with its agility in 1917, Fokker copied it to create its own Dr.I triplane as a mount for Baron Manfred von Richthofen.

The entry of the United States into the war in April 1917 inspired Germany with a desperate urge to achieve victory before the overwhelming might of American manpower and industry could be brought to bear. The German high command planned an ultimate offensive on the Western Front for the spring of 1918. Evaluation trials were held to find the aircraft that would win the war. Both Claudius Dornier and Hugo Junkers put forward radical designs that looked to the future – aircraft with metal skins and, in Junkers' case, a cantilever wing that required no external struts or bracing. But again conservatism prevailed. The aircraft adopted for mass production were the Fokker D.VII and Pfalz D.XII, superb fighting machines but representing only a limited degree of innovation.

Racing to catch up

The expectation that America's entry into the conflict would swiftly swing the air war in the Allies' favor failed to take into account the degree to which American aviation had fallen behind that of Europe. The sole viable combat aircraft under

Bristol F.2B Fighter

ALONG WITH THE S.E.5A AND THE SOPWITH CAMEL, the two-seater Bristol Fighter belonged to a new generation of British aircraft delivered to the Western Front in 1917. Designed as a reconnaissance fighter, with the second crew member acting as gunner and observer, it also served as a bomber and ground attack aircraft. It survived an inauspicious combat debut when, on April 5, 1917, six Bristol fighters from 48 Squadron RFC led by Captain Leece Robinson VC, ran into a formation of Albatroses commanded by Manfred von Richthofen. Four Bristol F.2A Fighters were shot down, including Leece Robinson's. They had been warned about alleged structural frailty, so had handled it too gingerly during the combat. However, pilots soon learned to handle it like a true fighter, using the forward-firing gun as the main armament, and figuring out the best formations to bring the rear gunner into play.



BARNWELL'S BRISTOL

The Bristol Fighter was designed by Captain Frank Sowter Barnwell of the British Colonial Aeroplane Company. It made its maiden flight in September 1916.

Fabric-covered wire-spoked wheels